

IMO(44th) – 2003

First Day

1. S is the set $\{1, 2, \dots, 1000000\}$. Show that for any subset A of S with 101 elements we can find 100 distinct elements x_i of S , such that the sets $\{a + x_i \mid a \in A\}$ are all pairwise disjoint.
2. Find all pairs (m, n) of positive integers such that $\frac{m^2}{2mn^2 - n^3 + 1}$ is a positive integer.
3. A convex hexagon has the property that for any pair of opposite sides the distance between their midpoints is $\sqrt{3}/2$ times the sum of their lengths. Show that all the hexagon's angles are equal.

Second Day

4. ABCD is cyclic. The feet of the perpendicular from D to the lines AB, BC, CA are P, Q, R respectively. Show that are angle bisector of ABC and CDA meet on the line AC iff $RP = RQ$.
5. Given $n > 2$ reals $x_1 \leq x_2 \leq \dots \leq x_n$, show that $\left(\sum_{i,j} |x_i - x_j| \right)^2 \leq \frac{2}{3} (n^2 - 1) \sum_{i,j} (x_i - x_j)^2$. Show that we have equality iff the sequence is an arithmetic progression.
6. Show that for each prime p , there exist a prime q such that $n^p - p$ is not divisible by q for any positive integer n .

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